Physics Based Simulation of a Three Dimensional LiDAR Sensor

MAE 5810 Robot Perception

Midterm Project Report

Author _____

Student ID _____

Instructor Silvia Ferrari

October 19 2026

Project evidence guide

Requirement	Report evidence
Physical model	Section 2; Figures 1, 3, 5
Virtual simulation	Section 3; Python implementation and saved runs
Dependencies and media	Section 4; Figures 1-7 and three videos
Technical analysis	Sections 4-5; model checks and limitations

Companion videos in simulalted_environments

environment_conditions.mp4; sensor_state_best_vs_worst.mp4; sampling_reflectance_factorial.mp4

Abstract

This project models a spinning three dimensional pulsed LiDAR with opaque visibility, surface response, attenuation, and receiver noise. Analytic pose checks, a four-case sampling and reflectance experiment, nine target settings, and fixed-scene weather navigation demonstrate the measurement dependencies. Mounting angles remain unchanged in the sampling and reflectance comparison. Fog reduces observed endpoint cells from 73.7 to 47.1 percent, although all nine weather runs reach the goal. An independent 33,000-trial threshold experiment gives probability calibration RMS error 0.55 percentage points. Figures, recorded histories, and three videos connect the model to simulation results. These checks establish internal consistency; physical calibration and general navigation reliability remain untested.

1 Introduction

A virtual LiDAR must determine where its beams can reach and whether their returned signal is measurable. A missing point may arise from angular sampling, occlusion, a range gate, or weak power. This project therefore models measurements together with field of view (FOV) and line of sight (LOS), addressing the dependencies on sensor state, targets, and environment required by the midterm assignment [1].

The navigation sensor uses eight channels. Surface modeling is motivated by Heiden et al. [2], whose continuous wave phase estimator differs from this pulsed geometric approximation. Pose, sampling, target, and weather comparisons isolate different mechanisms. Threshold trials resolve uncertain detections independently of navigation.

2 Theory and Methods

2.1 Scan pattern

The sensor frame has x forward, y left, and z up. World z is up. With M azimuth samples per full revolution and L channels, beam angles and the nominal unit direction are

$$\alpha_i = \alpha_{min} + \frac{2\pi i}{M}, \qquad \beta_j = \beta_{min} + \frac{j(\beta_{max} - \beta_{min})}{L - 1} \tag{1}$$

$$b_{ij} = [\cos \beta_j \cos \alpha_i, \cos \beta_j \sin \alpha_i, \sin \beta_j]^T \tag{2}$$

A full sweep uses i = 0, …, M − 1. A partial window uses the same azimuth spacing. A single channel takes the configured upper elevation. Angles in equations are in radians; plots use degrees.

2.2 Pose and coverage

All channels at one azimuth share a firing time. The origin translates at constant world velocity; yaw may change during a scan while roll and pitch remain fixed:

$$t_i = \frac{\alpha_i - \alpha_{min}}{2\pi f_{rot}}, \qquad o_i = o_0 + v t_i \tag{3}$$

The sensor to world rotation is R = Rz(yaw) Ry(pitch) Rx(roll), and the world beam is d = Rb. For point p, let q = R^T(p − o), r = ‖q‖, a = atan2(qy, qx), and e = arcsin(qz/r). The nominal FOV is

$$F = \{ p : r_{min} \le r \le r_{max},\ a \in A,\ \beta_{min} \le e \le \beta_{max} \} \tag{4}$$

A is the circular azimuth window. Discrete beams can miss targets within this continuous set. For each subray, the nearest opaque intersection with scene surfaces S is

$$r_k = min \{ s > 10^{-6}\ m : o + s d_k \in S \} \tag{5}$$

The first hit is found before range gating, so a weak or too near object still blocks farther geometry. A finite beam uses normalized Gaussian weighted subrays in the plane normal to its central direction. For full divergence δ, the angular standard deviation is δ/4; a sunflower pattern truncated at 2.5 standard deviations supplies the sample offsets.

2.3 Surface power and echo formation

For a ray facing normal n, μ = max(0, −n^Td). Diffuse weight ρd, specular weight ρs, and exponent m define the approximate surface law

$$f = \frac{\rho_d}{\pi} + \frac{\rho_s (m + 2)}{2\pi} [max(2\mu^2 - 1,\ 0)]^m \tag{6}$$

$$P_k = w_k P_t A_r \eta\ f \frac{\mu_k}{r_k^2}\ \exp(-2\kappa r_k) \tag{7}$$

Here Ar = π(Dr/2)², η is optical efficiency, and κ = κair + arain q_{rain}^b. Invalid hits contribute zero power. Rain and fog change attenuation, without backscatter returns. Consecutive subray hits separated by at most the pulse gate merge into an echo group G:

$$\Delta r_{gate} = \frac{c \tau_{FWHM}}{2}, \qquad P_G = \Sigma P_k, \qquad \bar{r}_G = \frac{\Sigma P_k r_k}{\Sigma P_k} \qquad (k \in G) \tag{8}$$

The centroid approximates pulsed ranging rather than waveform processing; it can lie between surfaces at edges. Geometry supports planes, spheres, yawed boxes, and capped cylinders.

2.4 Detection and range uncertainty

For an existing echo, noisy power is $\tilde{P} = PG + \nu$, with $\nu \sim N(0, \sigma P^2)$. Detection requires $\tilde{P} \geq \gamma\sigma P$, giving the single echo probability

$$P_D(G) = \Phi\left(\frac{P_G}{\sigma_P} - \gamma\right) \tag{9}$$

Φ is the standard normal cumulative distribution function. The strongest mode selects the greatest noisy detected power; first and last modes select by range. There is at most one retained return per nominal beam. With $S = \tilde{P}/\sigma P$, the assumed timing inspired range law is

$$\sigma_r^2(S) = \sigma_0^2 + \frac{(c\,t_{rise}/2)^2}{2\ max(S,\ 10^{-9})}, \qquad z = \bar{r}_G + \sigma_r(S)\xi \tag{10}$$

ξ is independent standard normal noise. This law is an assumption, not a derived Cramér–Rao bound. Selection truncates measured power but leaves the subsequent standardized range noise normal. Coordinates use the central beam and known firing pose:

$$\hat{p}^S = z\ b, \qquad \hat{p}^W = o + z\ d \tag{11}$$

2.5 Mathematical dependencies

A forward plane intersection is $r = n^T(p0 - o)/(n^Td)$. Horizontal ground at mounting height h and downward world beam elevation β therefore gives

$$r = -\frac{h}{\sin\ \beta}, \qquad \frac{\partial r}{\partial h} = -\frac{1}{\sin\ \beta}, \qquad \frac{\partial r}{\partial \beta} = \frac{h\ \cos\ \beta}{\sin^2\ \beta} \tag{12}$$

Height and orientation change range, with increasing sensitivity near horizontal beams. The FOV transformation changes coverage with pose; the nearest hit rule changes LOS at occluder edges. For a beam filling diffuse target, $P = C \exp(-2\kappa r)/r^2$ and $C = PtAr\eta\rho d \cos\theta/\pi$, so

$$\frac{\partial\ \ln\ P}{\partial r} = -\frac{2}{r} - 2\kappa, \qquad \frac{\partial\ \ln\ P}{\partial \kappa} = -2r, \qquad \frac{\partial\ \ln\ P}{\partial \theta} = -\tan\ \theta \tag{13}$$

Diffuse weight scales power linearly. Greater range, extinction, and incidence angle weaken the diffuse signal, and therefore reduce detection probability. The angle derivative holds range fixed and excludes the specular branch. For any selected physical reference,

$$E[(z - r_{ref})^2 \mid \tilde{P},\ G] = (\bar{r}_G - r_{ref})^2 + \sigma_r^2(S) \tag{14}$$

The first term is centroid displacement; the second is conditional random variance. Angular, pose, calibration, and additional physical model errors are excluded.

3 Procedure

Python and NumPy generate measurements; independent renderers read saved results. The fixed scene has 12 objects: nine in the planning map and three unknown to the planner. A diffuse study box is centered at (11, 1.1, 0.6) m with size (0.8, 1.6, 1.2) m and baseline weight 0.30.

Defaults are eight channels (-20 to +10 degrees), 360 azimuths, one subray, level mount at 0.6 m, 0.5-200 m gates, 75 W peak power, 25 mm aperture, 0.5 efficiency, 4×10^{-9} W power noise, threshold factor 10, 0.01 m range floor, 5 ns pulse, and 1 ns rise time.

A* plans from (0, 0) to (18, 0) m on a 0.25 m grid with 0.90 m inflation. A label-free measured-point controller uses $0.12 \le z \le 1$ m; limits are 1.2 m/s and 90 degrees/s. At 0.2 s steps, up to 36 s,

$$p_{k+1} = p_k + v_k \Delta t \, [\cos \theta_k, \sin \theta_k]^T, \qquad \theta_{k+1} = wrap(\theta_k + \omega_k \Delta t) \tag{15}$$

3.1 Sensor state and sensor configuration

State means position and orientation; sampling and noise are configuration parameters. A ground-height check changes position at fixed angles. The new factorial combines M = 360 or 90 with target diffuse weight 0.85 or 0.00005, keeping mounting angles, environment, geometry, route, and control settings fixed. The weak weight is a virtual stress setting. Each case has three navigation seeds and matched baseline-pose replays. A separate noise case increases σP tenfold.

3.2 Target characteristics

Nine settings vary diffuse weight (0.02, 0.30, 0.85), yaw (0, 45, 75 degrees), height (0.3, 1.2, 2.0 m), and center x (9, 11, 13 m). Reuse the same 271 baseline snapshots across three seeds. Count intersections, visible hits, in-gate eligible echoes NE, and detections ND separately. For qi in Equation (9),

$$\hat{p}_D = \frac{N_D}{N_E}, \qquad \bar{p}_D = \frac{\Sigma q_i}{N_E} \qquad (i \in E) \tag{16}$$

3.3 Environmental conditions through navigation

With obstacles, target, sensing, controller, and reference fixed, compare normal air ($\kappa = 0.0002$ m^{-1}), rain at 50 mm/h ($\kappa = 0.104764$ m^{-1}), and fog ($\kappa = 0.6$ m^{-1}), each at seeds 1-3. Retain all nine closed-loop histories and matched-pose replays. Points per scan include all surfaces; maps count observed endpoint cells. Target range RMS excludes misses, which remain in the detection denominator. Weather is attenuation only, without backscatter. Three seeds are descriptive.

4 Results

4.1 Target characteristics

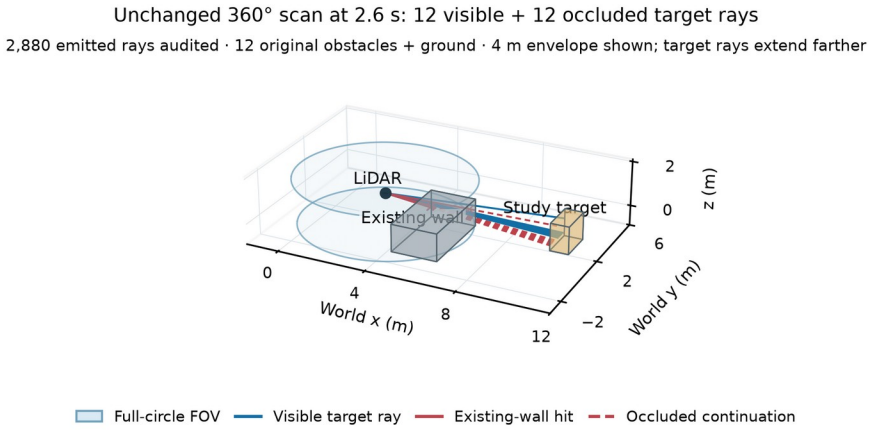


Figure 1 Unchanged baseline at 2.6 s: 2,880 emitted rays; 12 visible and 12 existing-wall-blocked target rays. Only representative rays and relevant true surfaces are drawn. The full-circle angular envelope is clipped at 4 m; target rays extend farther. This is explanatory geometry, not measured map points.

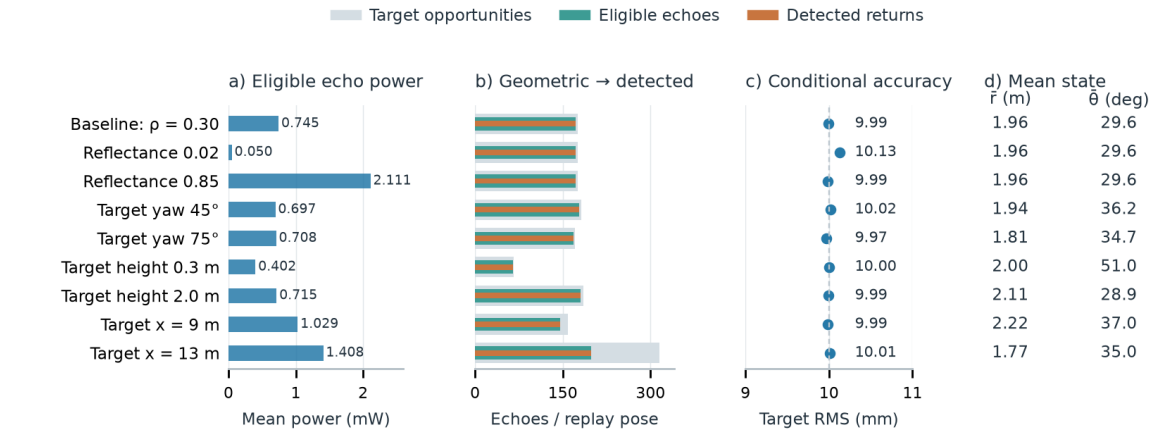


Figure 2 All nine target settings at matched baseline poses. Power is the noiseless eligible-echo mean; counts distinguish intersections, eligibility, and detections. Range RMS is conditional on detected target returns. Target position, orientation, and size also change geometric exposure.

Diffuse weights 0.02, 0.30, and 0.85 give power ratios 0.0667, 1, and 2.8333. At one common seed-1 viewpoint (6 s), target centers x = 9, 11, and 13 m give ranges 2.134, 4.011, and 5.950 m and powers 0.339830, 0.105144, and 0.048696 mW. Whole-route averages need not preserve this ordering because eligible exposures and incidence differ.

4.2 Sensor state and sensor configuration

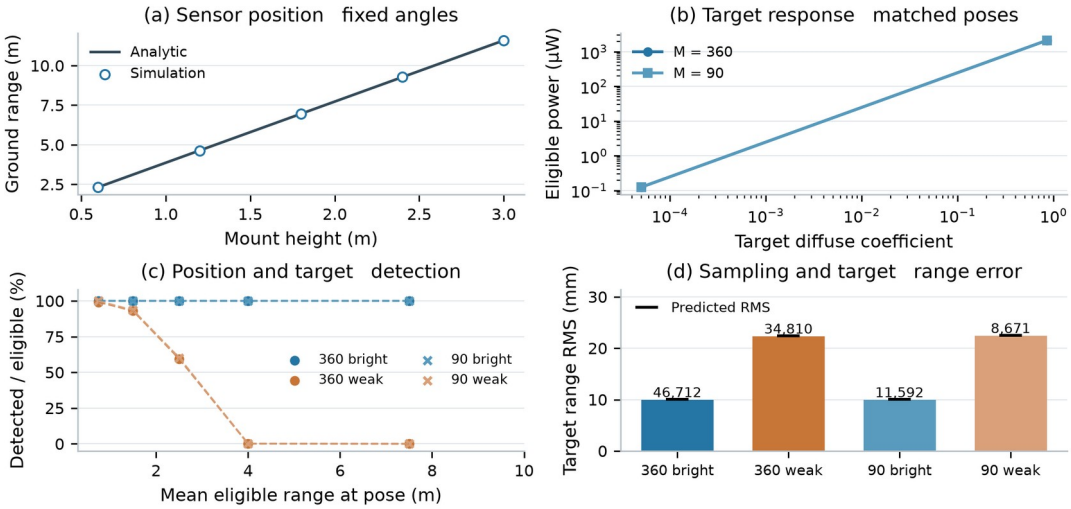


Figure 3 (a) Ground range versus mounting height at fixed angles. (b-d) Four sampling/reflectance combinations at matched poses. Dashed curves and black markers are predictions; dots and bars are simulations. Detection bins group poses by mean eligible range; RMS labels give detection counts.

Height 0.6-3 m gives ground ranges 2.318-11.591 m at fixed -15 degree elevation. Separately, an 80 degree FOV at yaw 0/40/45/50 degrees gives 132/70/6/0 target returns; full-circle integer-grid yaw retains 132. Both state components matter; factorial mounting angles stay fixed.

Bright/weak power scales by 17,000. Fourfold beam sampling gives 46,712/11,592 bright-target detections; grids sample different features. Dense weak/bright detection is 74.52/100 percent and RMS 22.34/9.99 mm. Tenfold receiver noise changes scene returns 2,198.4 → 2,011.2 per scan and target RMS 9.992 → 10.082 mm; target detection stays saturated.

4.3 Environmental measurement metrics

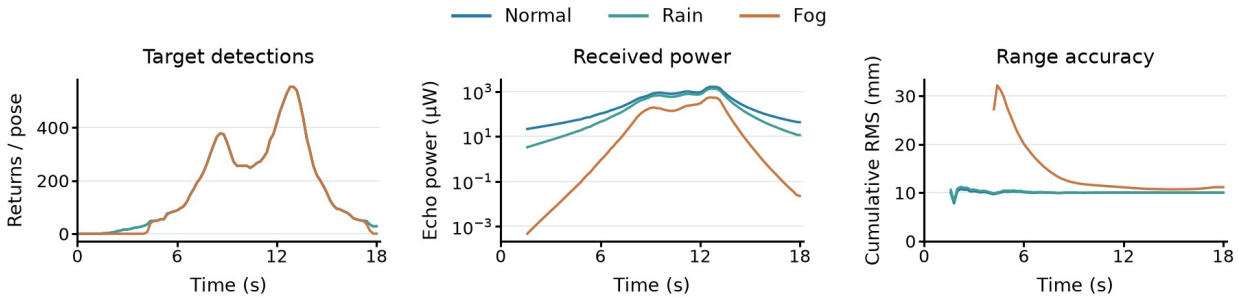


Figure 4 Normal, rain, and fog at matched baseline poses, all seeds. Cumulative target RMS uses actual detections. Attenuation reduces power; the range floor limits RMS changes for strong returns. Profiles stop at recorded exposure.

4.4 Near threshold detection and range uncertainty

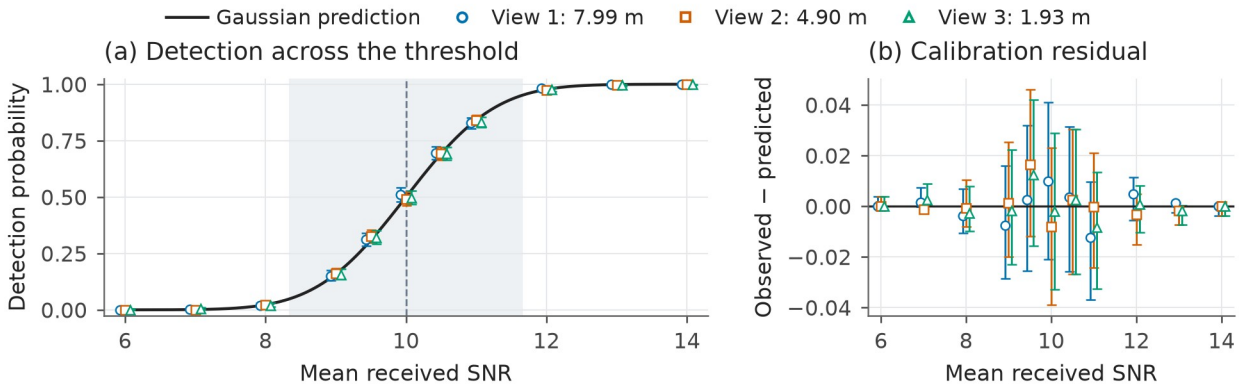


Figure 5 Detection and calibration residuals from 33,000 actual scan trials. Each marker counts 1,000 trials, including misses; bars are pointwise 95% Wilson intervals. Small horizontal offsets distinguish viewpoints. The shaded band is the predicted 5-95% transition, not an uncertainty interval.

The weather replays nearly saturate detection. For independent component validation, select three eligible rays from this same baseline scene at ranges 7.99, 4.90, and 1.93 m. Vary extinction to obtain 11 mean-SNR settings from 6 to 14, with 1,000 independent actual scans per setting. Geometry, diffuse target response, receiver noise, and threshold remain fixed. Ray selection does not depend on noisy detection; these pencil-ray probes are not additional navigation runs.

At mean SNR 10, predicted probability is 0.5; measured fractions are 0.510, 0.492, and 0.498. Over 33 settings, calibration RMS error is 0.547 percentage points, with maximum residual 1.646 points. The model lies within 32/33 pointwise intervals; the excluded prediction is retained.

With n independent trials, observed fraction $\hat{p}$, and a = 1.96, the Wilson interval is

$$\frac{\hat{p} + \frac{a^2}{2n} \pm a\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{a^2}{4n^2}}}{1 + \frac{a^2}{n}} \tag{17}$$

For detected echoes only, define range and standardized errors; misses are not zero errors:

$$e_i = z_i - r_{true,i}, \qquad \varepsilon_i = \frac{e_i}{\sigma_r(S_i)}, \qquad RMSE = \sqrt{\frac{\Sigma e_i^2}{N_D}} \tag{18}$$

Across 16,507 detections, range RMS is 32.358 mm versus 32.287 mm predicted from measured-SNR conditional variance; standardized RMS is 1.0018. Agreement checks the assumed noise and selection laws, not physical accuracy. Intervals do not provide simultaneous coverage.

4.5 Environmental conditions through navigation

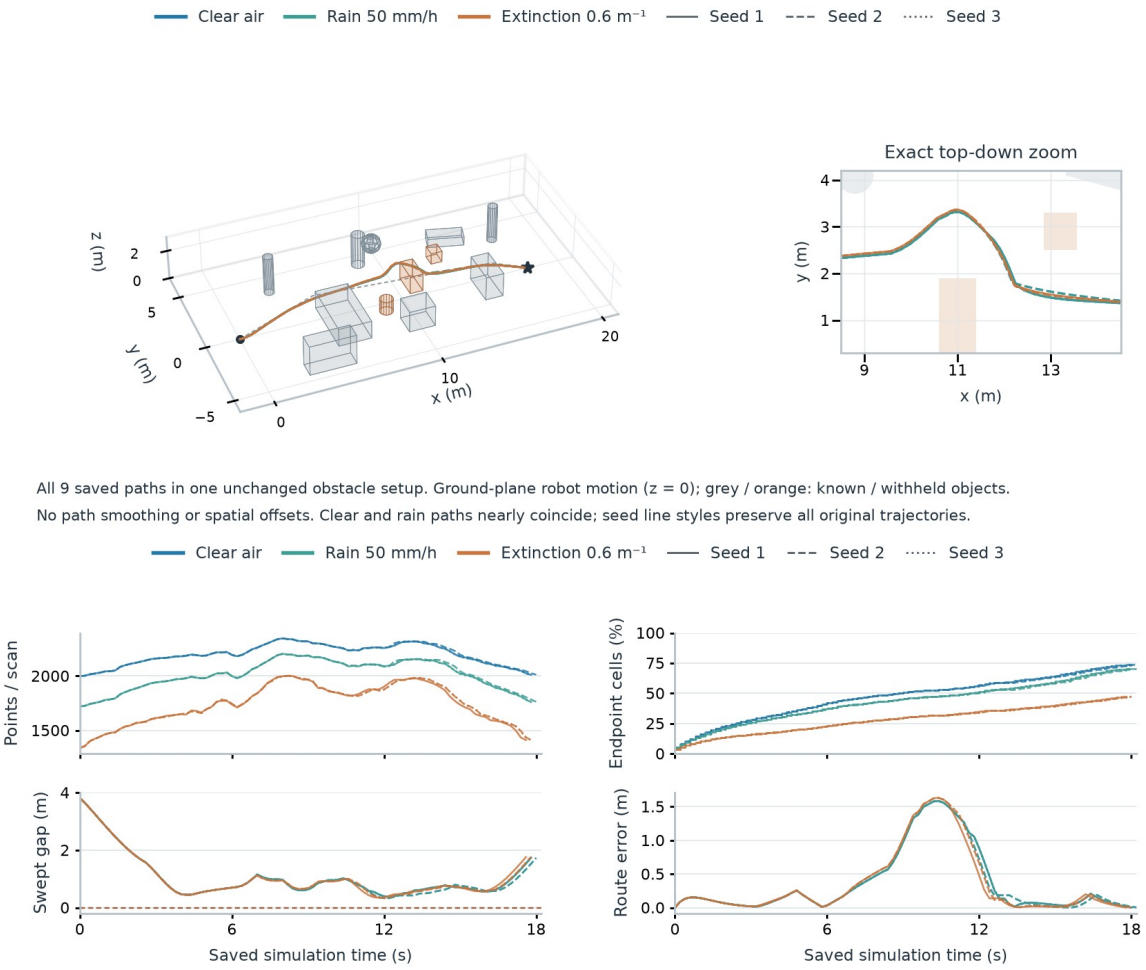


Figure 6 All nine weather runs in the unchanged scene. Upper: recorded 3D paths and top-down detail. Lower: points per scan, endpoint coverage, swept clearance, and route error. Exact overlaps remain overlaps. Lines stop at recorded termination; terminal poses add no scan. Colors denote weather, line styles seeds.

Table 1 Three-seed means, except pooled target RMS and worst swept gap. Map counts observed endpoint cells. All runs reach the goal without collision or timeout; three seeds are descriptive.

Condition	Points / scan	Map (%)	RMS (mm)	Goals	Time (s)	Path (m)	Gap (m)
Normal	2,198.4	73.67	9.992	3/3	18.07	19.702	0.330
Rain 50 mm/h	2,011.3	70.33	10.006	3/3	18.07	19.702	0.330
Fog $\kappa = 0.6\ m^{-1}$	1,742.0	47.10	11.139	3/3	17.93	19.784	0.345

4.6 Measured mapping and media evidence

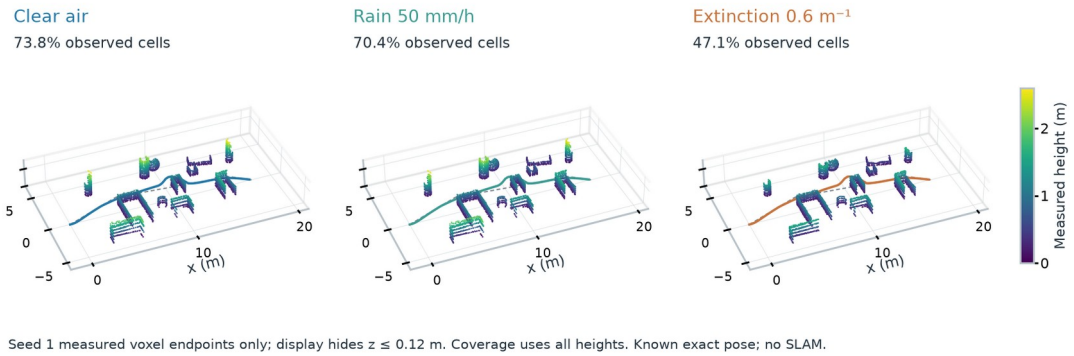


Figure 7 Seed-1 normal, rain, and fog maps; shared axes and height colors. Display hides z ≤ 0.12 m. Scene truth never fills gaps.

Normal, rain, and fog detect 100, 100, and 98.07 percent of eligible target echoes. Fog reduces scene returns by 20.8 percent and endpoint coverage by 26.57 percentage points. All weather runs still reach the goal; map observations do not establish accuracy or free space.

Maps use 0.12 m voxels and 0.2 m endpoint cells over [-2, 21] m × [-6, 7] m (NROI = 7,475). For K intervals,

$$L = \Sigma \| p_{k+1} - p_k \|, \qquad f_{obs} = \frac{N_{obs}}{N_{ROI}}, \qquad k = 0, \ldots, K - 1 \tag{19}$$

The companion videos show weather, dense/bright versus sparse/weak settings, and all four sampling/reflectance combinations. Figure 3 isolates measurement causes. The favorable/adverse pair does not establish global optima or guaranteed navigation outcomes.

5 Conclusions

Residuals are at most 3.83e-13 m in 65 geometry checks and 4.39e-16 in 72 power checks. Controlled dependencies and threshold calibration support internal consistency, not physical accuracy or navigation reliability. Backscatter, pose uncertainty, moving targets, and closed-loop scan motion remain untested. Mapping uses exact poses, not SLAM.

References

[1] “Midterm project,” Robot Perception, course assignment handout, pp. 1-2.

[2] E. Heiden, Z. Liu, R. K. Ramachandran, and G. S. Sukhatme, “Physics-based Simulation of Continuous-Wave LIDAR for Localization, Calibration and Tracking,” Proc. IEEE ICRA, 2020, pp. 2595-2601.